Mathematics Advanced | Year 12 | MA-M1 Modelling Financial Situations

Teacher Teaching Guide

Applications of Arithmetic and Geometric Sequences

等差数列与等比数列的应用 | 教师教学指南

Key words with Chinese translations • NESA-style worked solutions • 10 HSC-style practice questions

Learning intentions

Students are learning to:

- recognise from the wording of a problem whether it describes an arithmetic sequence (AP) or a geometric sequence (GP)
- identify a, d or r, and n from key words, and choose the correct formula from the NESA Reference Sheet
- solve applied problems, including simple interest (AP) and compound interest, depreciation and annual deposits (GP, compounded annually)

Success criteria

I can:

- highlight key words and turn them into a, d, r and n
- decide whether a question needs a term T_n, a sum S_n or a limiting sum S
- solve "first exceeds" and "first falls below" problems using algebra or logarithms, and check my answer
- write a full NESA-style solution that ends with a statement in context (units, rounding, whole numbers)

Syllabus links: MA-M1.2 Arithmetic sequences and series; MA-M1.3 Geometric sequences and series; MA-M1.4 Financial applications. **Outcomes:** MA12-2, MA12-4, MA12-10.

How this guide is organised

Section	Content	Suggested time
A	Key words for AP and GP, with Chinese translations	15 min
B	Worked examples with key words highlighted: 3 AP and 3 GP (no finance)	45 min
C	GP in finance: (i) key words and vocabulary; (ii) 3 examples (compounded annually only)	40 min
D	AP in finance: (i) key words (simple interest, constant increases); (ii) 3 examples	30 min
E	10 mixed HSC-style practice questions (easy, moderate, challenging), then NESA-style solutions	60 min or homework
–	Vocabulary list for AP and GP applications, with Chinese translations (last page)	reference

Formulas used in this guide (all from the NESA Mathematics Advanced Reference Sheet)

Sequences and series

AP: $T_n = a + (n-1)d$ $\quad S_n = \dfrac{n}{2}[2a + (n-1)d] = \dfrac{n}{2}(a + l)$

GP: $T_n = ar^{n-1}$ $\quad S_n = \dfrac{a(1-r^n)}{1-r} = \dfrac{a(r^n-1)}{r-1},\ r \neq 1$ $\quad S = \dfrac{a}{1-r},\ |r| < 1$

Financial Mathematics $A = P(1+r)^n$ $\quad$ **Logarithms** $\log_a a^x = x = a^{\log_a x}$, $\quad \log_a x = \dfrac{\log_b x}{\log_b a}$

Quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The simple interest formula is *not* on the Advanced Reference Sheet. In this guide, simple interest is modelled as an arithmetic sequence: the interest earned each year is the same amount, so it is the common difference d.

A problem-solving routine for every question

1. Read & highlight — Find the key words (Section A) → **2. AP or GP?** — add the same amount ⇒ AP; multiply by the same factor ⇒ GP → **3. Write a, d or r, n** — and say what n counts (rows, days, years) → **4. Formula & answer** — use the Reference Sheet, then answer in context

Section A

Key Words for Applications of AP and GP

关键词：等差数列与等比数列的应用题

A1 Key words that signal an arithmetic sequence (AP) 等差数列

An AP changes by **adding or subtracting the same amount** each time. This amount is the common difference, d.

AP key words and phrases	中文翻译 Chinese	What it tells you
increases by 3 seats every row	每排增加 3 个座位	Add the same amount each time: AP with $d = 3$ (20, 23, 26, …)
each row has 2 more tiles than the previous row	每一排比前一排多 2 块瓷砖	$d = 2$ (3, 5, 7, …)
decreases by 5 each week; 5 fewer each week	每周减少 5；每周少 5 个	$d = -5$ (80, 75, 70, …)
goes up by the same amount each time	每次增加相同的数量	Constant difference, so it is an AP
a constant (fixed) increase or decrease	固定的增加量或减少量	The common difference d
the first row has …; initially; starts with	第一排有……；最初；开始时	The first term $a = T_1$
in the 20th row; on day n	在第 20 排；在第 n 天	One term: T_{20}, T_n
altogether; in total; the total of the first n	一共；总共；前 n 个的总和	A sum: S_n
the last row has …	最后一排有……	The last term l
how many rows are needed for a total of …	需要多少排才能使总数达到……	Solve S_n = total for n
first exceeds; first more than	首次超过；首次多于	Solve, then round **up** to the next whole number
at most; maximum capacity; cannot exceed	最多；最大容量；不能超过	Solve, then round **down**
successive; consecutive	相继的；连续的	Terms that follow one after another
linear growth; linear decay	线性增长；线性减少	Changes by a constant amount: AP

A2 Key words that signal a geometric sequence (GP) 等比数列

A GP changes by **multiplying by the same number** each time. The number is the common ratio r. A percentage increase or decrease is a GP.

GP key words and phrases	中文翻译 Chinese	What it tells you
doubles every hour	每小时翻一倍	Multiply by 2 each time: GP with $r = 2$ (500, 1000, 2000, …)
triples each day	每天变为原来的三倍	$r = 3$
halves each time	每次减半	$r = \frac{1}{2}$
increases by 4% each year	每年增加 4%	$r = 1 + 0.04 = 1.04$ (*not* "add 4")
decreases by 20% each week	每周减少 20%	$r = 1 - 0.20 = 0.8$
rebounds to 75% of its previous height	反弹到前一次高度的 75%	$r = 0.75$
each term is a fixed multiple of the previous term	每一项都是前一项的固定倍数	Constant ratio, so it is a GP
grows exponentially	呈指数增长	GP with $r > 1$
decays exponentially	呈指数衰减	GP with $0 < r < 1$
after n hours (starting value at time 0)	n 小时之后（从时间 0 开始）	The $(n+1)$th term: $T_{n+1} = ar^n$
in the nth week	在第 n 周	The nth term: $T_n = ar^{n-1}$
eventually; in the long run; sum to infinity	最终；长期来看；无穷项的和	Limiting sum $S = \frac{a}{1-r}$, $\lvert r\rvert < 1$
will never exceed; can never reach	永远不会超过；永远达不到	Compare with the limiting sum
first falls below	首次低于	Solve using logarithms, then round up

A3 AP or GP? A quick test

	Arithmetic sequence (AP) 等差数列	Geometric sequence (GP) 等比数列
How it changes	adds or subtracts the same amount	multiplies by the same number
Test	$T_2 - T_1 = T_3 - T_2 = d$	$\frac{T_2}{T_1} = \frac{T_3}{T_2} = r$
Typical words	"2 more", "5 fewer", "increases by \$100"	"doubles", "halves", "increases by 4%", "80% of"
Graph of T_n	points lie on a straight line (linear)	points lie on a curve (exponential)

Teaching notes: common mistakes

- "Increases by 4%" means multiply by 1.04. It does not mean add 4 or multiply by 0.04.
- If the starting value is at time 0, the value "after n years" is the $(n+1)$th term, but the value "in the nth year" is the nth term.
- "First exceeds" and "first falls below": solve the equation, round **up**, then check the two whole numbers either side. "At most": round **down**.
- Reject answers that do not make sense in context, such as negative rows or a fraction of a day.

Section B

Worked Examples: Key Words in Context

3 AP examples and 3 GP examples (no finance) | 例题：在题目中找出关键词

Key words are **highlighted** in each question. The **Key words** box shows how each phrase turns into a, d, r or n. Model this with the class: read the question aloud, highlight together, then fill in the key words box before solving.

Example B1 AP Easy — Theatre seating

The first row of a theatre has 20 seats. The number of seats **increases by 3 seats every row**, and there are 25 rows.

(a) How many seats are in **the last row**? (1 mark)

(b) How many seats are there in the theatre **altogether**? (2 marks)

Key words 关键词

increases by 3 seats every row （每排增加 3 个座位）	⇒ same amount added each row: AP with $d = 3$
first row has 20 seats （第一排有 20 个座位）	⇒ first term $a = 20$
the last row （最后一排）	⇒ the 25th term, T_{25}
altogether （一共）	⇒ the sum of all 25 rows, S_{25}

NESA-style solution

(a) The numbers of seats in the rows form an arithmetic sequence with $a = 20$ and $d = 3$.
Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{25} = 20 + (25-1) \times 3 = 20 + 72 = 92$$

$\therefore$ There are 92 seats in the last row.

(b) Using $S_n = \frac{n}{2}(a + l)$ (Reference Sheet) with $n = 25$, $a = 20$ and $l = 92$:

$$S_{25} = \frac{25}{2}(20 + 92) = \frac{25}{2} \times 112 = 1400$$

$\therefore$ There are 1400 seats altogether.

Teaching notes

Either sum formula works. $S_n = \frac{n}{2}[2a + (n-1)d]$ gives the same answer without finding the last term first.

Example B2 AP Moderate — Training for a fun run

Mia is training for a fun run. She runs 2 km on the first day and **increases her distance by 0.5 km each day**.

(a) How far does she run **on the 15th day**? (1 mark)

(b) On which day does she **first run more than** 10 km? (2 marks)

(c) Find the **total distance** she runs in the **first 4 weeks**. (2 marks)

Key words 关键词

increases her distance by 0.5 km each day （每天增加 0.5 公里）	⇒ AP with $d = 0.5$
2 km on the first day （第一天跑 2 公里）	⇒ $a = 2$
on the 15th day （在第 15 天）	⇒ one term, T_{15}
first run more than （首次跑超过）	⇒ solve $T_n = 10$, then take the next whole day
total distance （总距离）	⇒ a sum, S_n
first 4 weeks （前 4 周）	⇒ $n = 4 \times 7 = 28$ days

NESA-style solution

(a) The daily distances form an arithmetic sequence with $a = 2$ and $d = 0.5$.
Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{15} = 2 + 14 \times 0.5 = 9$$

$\therefore$ She runs 9 km on the 15th day.

(b) $T_n = 2 + (n-1)(0.5) = 0.5n + 1.5$. Solve $T_n = 10$:

$$0.5n + 1.5 = 10 \quad \Longrightarrow \quad 0.5n = 8.5 \quad \Longrightarrow \quad n = 17$$

On day 17 she runs exactly 10 km, which is *not more than* 10 km. Her distance increases every day, so the first day she runs more than 10 km is day 18 ($T_{18} = 10.5$ km).
$\therefore$ Day 18.

(c) 4 weeks is 28 days. Using $S_n = \dfrac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_{28} = \frac{28}{2}[2(2) + 27(0.5)] = 14 \times 17.5 = 245$$

$\therefore$ She runs a total of 245 km in the first 4 weeks.

Teaching notes

"More than" is strict. If the equation gives a whole number, check whether that term equals the target. If it does, the answer is the next term.

Example B3 AP Challenging — Stacking cans

A supermarket stacks cans in rows. The bottom row has 40 cans and **each row has 2 fewer cans than the row below it**.

(a) Show that the **total number of cans** in the bottom n rows is $n(41 - n)$. (2 marks)

(b) The display uses **exactly 400 cans**. How many rows does the display have, and how many cans are in the top row? (3 marks)

Key words 关键词

2 fewer cans than the row below （比下面一排少 2 个罐头）	⇒ AP with $d = -2$ (counting up from the bottom)
bottom row has 40 cans （最底排有 40 个罐头）	⇒ $a = 40$
total number of cans （罐头总数）	⇒ a sum, S_n
exactly 400 cans （正好 400 个罐头）	⇒ solve $S_n = 400$

NESA-style solution

(a) Counting from the bottom, the rows form an arithmetic sequence with $a = 40$ and $d = -2$.
Using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_n = \frac{n}{2}[2(40) + (n-1)(-2)] = \frac{n}{2}(82 - 2n) = n(41 - n) \quad \text{as required.}$$

(b) Solve $S_n = 400$:

$$\begin{aligned} n(41 - n) &= 400 \\ n^2 - 41n + 400 &= 0 \\ (n - 16)(n - 25) &= 0 \quad \Longrightarrow \quad n = 16 \text{ or } n = 25 \end{aligned}$$

If $n = 25$, the top row would have $T_{25} = 40 + 24(-2) = -8$ cans, which is impossible. So $n = 16$.
Top row: $T_{16} = 40 + 15(-2) = 10$ (using $T_n = a + (n-1)d$ (Reference Sheet)).
$\therefore$ The display has 16 rows, and the top row has 10 cans.

Teaching notes

A sum can lead to a quadratic equation with two solutions. Always check which solution makes sense in the context.

Example B4 GP Easy — Bouncing ball

A ball is dropped from a height of 12 m. After each bounce it **rebounds to three-quarters of its previous height**.

(a) Find the height the ball reaches **after the 4th bounce**, correct to 2 decimal places. (2 marks)

(b) The ball **eventually** comes to rest. Find the **total vertical distance** it travels. (2 marks)

Key words 关键词

rebounds to three-quarters of its previous height （反弹到前一次高度的四分之三）	⇒ GP with $r = \frac{3}{4}$
after the 4th bounce （第 4 次反弹后）	⇒ the 4th term of the rebound heights
eventually （最终）	⇒ use the limiting sum
total vertical distance （总垂直距离）	⇒ add every fall and every rise

NESA-style solution

(a) The rebound heights $9, 6.75, \ldots$ form a geometric sequence with $a = 12 \times \frac{3}{4} = 9$ and $r = \frac{3}{4}$.
Using $T_n = ar^{n-1}$ (Reference Sheet):

$$T_4 = 9 \times \left(\frac{3}{4}\right)^3 = 3.796875 \approx 3.80$$

$\therefore$ After the 4th bounce the ball reaches a height of 3.80 m.

(b) The ball first falls 12 m. After that, each rebound height is travelled twice (up, then down).
Since $|r| = \frac{3}{4} < 1$, the rebound heights have a limiting sum. Using $S = \dfrac{a}{1-r}$ (Reference Sheet):

$$S = \frac{9}{1 - \frac{3}{4}} = 36 \qquad \text{Total distance} = 12 + 2 \times 36 = 84$$

$\therefore$ The ball travels a total vertical distance of 84 m.

Teaching notes

Draw the bounce diagram. The first drop is counted once; every rebound height is counted twice.

Example B5 GP Moderate — Population growth

The population of a town is 25 000. The population is expected to **increase by 4% each year**.

(a) Show that the population **after n years** is $25\,000 \times 1.04^n$. (1 mark)

(b) Find the population after 10 years, to the nearest whole number. (1 mark)

(c) After how many full years will the population **first exceed** 50 000? (2 marks)

Key words 关键词

increase by 4% each year （每年增加 4%）	⇒ GP with $r = 1.04$
after n years （n 年之后）	⇒ the $(n+1)$th term (the starting population is term 1)
first exceed （首次超过）	⇒ solve with logarithms, then round up

NESA-style solution

(a) The populations at the start, after 1 year, after 2 years, … are $25\,000,\ 25\,000 \times 1.04,\ 25\,000 \times 1.04^2, \ldots$
This is a geometric sequence with $a = 25\,000$ and $r = 1.04$. After n years is the $(n+1)$th term.
Using $T_n = ar^{n-1}$ (Reference Sheet):

$$T_{n+1} = 25\,000 \times 1.04^{(n+1)-1} = 25\,000 \times 1.04^n \quad \text{as required.}$$

(b) $25\,000 \times 1.04^{10} = 37\,006.1\ldots$ $\quad\therefore$ The population after 10 years is 37 006.

(c) Solve $25\,000 \times 1.04^n = 50\,000$:

$$\begin{aligned} 1.04^n &= 2 \\ n &= \log_{1.04} 2 && (\log_a a^x = x \text{ (Reference Sheet)}) \\ &= \frac{\ln 2}{\ln 1.04} = 17.67\ldots && (\text{change of base (Reference Sheet)}) \end{aligned}$$

The population increases every year, so it first exceeds 50 000 after 18 full years.
Check: after 17 years, $25\,000 \times 1.04^{17} \approx 48\,698 < 50\,000$; after 18 years, $25\,000 \times 1.04^{18} \approx 50\,645 > 50\,000$.
$\therefore$ After 18 years.

Teaching notes

Both logarithm steps are on the Reference Sheet: $\log_a a^x = x$ gives $n = \log_{1.04} 2$, and the change of base formula lets students evaluate it as $\frac{\ln 2}{\ln 1.04}$ on a calculator.

Example B6 GP Challenging — A swinging pendulum

A pendulum swings through an arc of 50 cm on its first swing. **Each swing is 80% of the length of the previous swing**.

(a) Find the length of the **8th swing**, correct to 2 decimal places. (1 mark)

(b) Find the **total distance** travelled by the pendulum in the **first 8 swings**, correct to the nearest centimetre. (2 marks)

(c) Which swing is the **first to be less than** 5 cm? (2 marks)

(d) Explain why the pendulum will **never travel more than** 2.5 m in total. (2 marks)

Key words 关键词

each swing is 80% of the previous（每次摆动是前一次的 80%）	⇒ GP with $r = 0.8$
first swing of 50 cm（第一次摆动 50 厘米）	⇒ $a = 50$
the 8th swing（第 8 次摆动）	⇒ one term, T_8
total distance in the first 8 swings（前 8 次摆动的总距离）	⇒ a sum, S_8
first to be less than（首次小于）	⇒ solve with logarithms, then round up
never travel more than（永远不会超过）	⇒ use the limiting sum

NESA-style solution

(a) The swing lengths form a geometric sequence with $a = 50$ and $r = 0.8$.
Using $T_n = ar^{n-1}$ (Reference Sheet):

$$T_8 = 50 \times 0.8^7 = 10.48576 \approx 10.49 \qquad \therefore \text{ The 8th swing is 10.49 cm.}$$

(b) Using $S_n = \dfrac{a(1-r^n)}{1-r}$ (Reference Sheet):

$$S_8 = \frac{50\left(1-0.8^8\right)}{1-0.8} = 250\left(1-0.8^8\right) = 208.05\ldots \approx 208 \qquad \therefore 208 \text{ cm}$$

(c) Solve $T_n = 5$:

$$\begin{aligned} 50 \times 0.8^{n-1} &= 5 \\ 0.8^{n-1} &= 0.1 \\ n-1 &= \log_{0.8} 0.1 = \frac{\ln 0.1}{\ln 0.8} = 10.31\ldots \qquad \text{(change of base (Reference Sheet))} \\ n &= 11.31\ldots \end{aligned}$$

The swings get shorter each time, so the first swing less than 5 cm is the 12th swing.
Check: $T_{11} = 50 \times 0.8^{10} \approx 5.37$ cm; $T_{12} = 50 \times 0.8^{11} \approx 4.29$ cm.
$\therefore$ The 12th swing.

(d) Since $|r| = 0.8 < 1$, the series of swing lengths has a limiting sum. Using $S = \dfrac{a}{1-r}$ (Reference Sheet):

$$S = \frac{50}{1-0.8} = 250 \text{ cm} = 2.5 \text{ m}$$

The total distance gets closer and closer to 2.5 m but never reaches it. So the pendulum never travels more than 2.5 m.

Section C

Finance Applications of GP (Compounded Annually)

(i) Key words and vocabulary | (ii) 3 worked examples | 等比数列在金融中的应用（按年复利）

C(i) Key words and vocabulary for GP finance 复利、折旧与定期存款

With compound interest the balance is **multiplied by** $(1 + r)$ **every year**, so the yearly balances form a GP. Use $A = P(1 + r)^n$ (Reference Sheet) for one amount, and a GP sum for regular deposits.

Key words and vocabulary	中文翻译 Chinese	What it tells you
compound interest	复利	Interest is added to the balance, then earns interest itself: GP
compounded annually	按年复利（每年计息一次）	Interest is added once a year, so n = number of years
principal; initial investment	本金；初始投资	P, the starting amount
interest rate per annum (p.a.)	年利率	r as a decimal (6% p.a. $\Rightarrow r = 0.06$)
amount; future value	本利和；终值	A, the value after n years
invested for n years	投资 n 年	n in $A = P(1 + r)^n$
interest earned	所得利息	$A - P$
grows by 6% p.a.	每年增长 6%	Multiply by 1.06 each year: GP with ratio 1.06
depreciates by 15% p.a.	每年折旧（贬值）15%	Multiply by 0.85 each year; use $r = -0.15$ in $A = P(1 + r)^n$
reducing (declining) balance	余额递减法	Depreciation by a fixed % of the *current* value: GP
salvage value; book value	残值；账面价值	The value after depreciation
deposits \$3000 at the start of each year	每年年初存入 3000 元	Each deposit grows for a different number of years: sum of a GP
annuity; regular contributions	年金；定期供款	Equal payments made every period
superannuation	养老金（退休金）	Retirement savings with regular contributions
balance	余额	The money in the account at a given time
doubles in value	价值翻倍	Solve $A = 2P$

Teaching notes

- For depreciation, keep using the Reference Sheet formula $A = P(1 + r)^n$ with a *negative* rate, e.g. $r = -0.15$, so $1 + r = 0.85$.
- For regular deposits, write each deposit separately with $A = P(1 + r)^n$, then add them: the total is a GP sum.

C(ii) Worked examples: GP finance (compounded annually)

Example C1 GP • finance Easy Compound interest

Ethan **invests $5000** at **6% p.a.**, **compounded annually**.

(a) Show that the amounts in the account at the end of each year form a geometric sequence. (1 mark)

(b) Find the **amount** in the account after 8 years. (1 mark)

(c) How much **interest is earned** in 8 years? (1 mark)

Key words 关键词

invests $5000（投资 5000 元）	⇒ principal $P = 5000$
6% p.a.（年利率 6%）	⇒ $r = 0.06$
compounded annually（按年复利）	⇒ n counts years
amount（本利和）	⇒ A
interest is earned（所得利息）	⇒ $A - P$

NESA-style solution

(a) Each year the interest is added to the balance, so the balance is multiplied by $1 + 0.06 = 1.06$:

$$\text{end of year 1: } 5000 \times 1.06 = 5300, \qquad \text{end of year 2: } 5300 \times 1.06 = 5618, \qquad \ldots$$

The ratio of successive amounts is always 1.06 (e.g. $\frac{5618}{5300} = 1.06$), so the amounts form a geometric sequence with ratio 1.06.

(b) Using $A = P(1 + r)^n$ (Reference Sheet) with $P = 5000$, $r = 0.06$ and $n = 8$:

$$A = 5000(1 + 0.06)^8 = 5000(1.06)^8 = 7969.240\ldots$$

$\therefore$ The amount after 8 years is \$7969.24.

(c) Interest earned $= A - P = 7969.24 - 5000 = \$2969.24$

Example C2 GP • finance Moderate — Depreciation of a car

A car is bought for \$42 000. Its value **depreciates by 15% p.a.**

(a) Find the value of the car after 5 years. (1 mark)

(b) After how many years will its value **first fall below** \$10 000? (2 marks)

Key words 关键词

depreciates by 15% p.a. （每年折旧 15%）	⇒ multiply by 0.85 each year; $r = -0.15$ in $A = P(1+r)^n$
bought for \$42 000 （以 42 000 元购买）	⇒ $P = 42\,000$
first fall below （首次低于）	⇒ solve with logarithms, then round up

NESA-style solution

(a) Using $A = P(1+r)^n$ (Reference Sheet) with $P = 42\,000$, $r = -0.15$ and $n = 5$:

$$A = 42\,000(1-0.15)^5 = 42\,000(0.85)^5 = 18\,635.62\ldots$$

$\therefore$ The value after 5 years is \$18 635.62.

(b) Solve $42\,000(0.85)^n = 10\,000$:

$$0.85^n = \frac{10\,000}{42\,000} = \frac{5}{21}$$

$$n = \log_{0.85}\left(\frac{5}{21}\right) = \frac{\ln\left(\frac{5}{21}\right)}{\ln 0.85} = 8.83\ldots \qquad \text{(change of base (Reference Sheet))}$$

The value decreases every year, so it first falls below \$10 000 after 9 years.

Check: after 8 years, $42\,000(0.85)^8 \approx \$11\,444.60$; after 9 years, $42\,000(0.85)^9 \approx \$9727.91$.

$\therefore$ After 9 years.

Teaching notes

Depreciation is a GP with a ratio less than 1. Writing it as $A = P(1+r)^n$ with $r = -0.15$ keeps students on the Reference Sheet formula.

Example C3 GP • finance Challenging — Annual deposits (superannuation)

At the **start of each year**, Ava **deposits \$3000** into a superannuation account that earns **7% p.a., compounded annually**.

(a) Show that the amount in the account at the end of 3 years is $3000\left(1.07 + 1.07^2 + 1.07^3\right)$. (2 marks)

(b) Find the amount in the account at the end of 25 years. (2 marks)

(c) How much **interest** has the account earned in 25 years? (1 mark)

Key words 关键词

at the start of each year （每年年初）	⇒ each deposit earns interest for a different number of years
deposits \$3000 （存入 3000 元）	⇒ the regular contribution, $P = 3000$
7% p.a., compounded annually （年利率 7%，按年复利）	⇒ multiply by 1.07 for each year
interest （利息）	⇒ final amount minus total deposits

NESA-style solution

(a) Using $A = P(1 + r)^n$ (Reference Sheet) for each deposit, with $P = 3000$ and $r = 0.07$:

1st deposit (start of year 1) earns interest for 3 years:	$3000(1.07)^3$
2nd deposit (start of year 2) earns interest for 2 years:	$3000(1.07)^2$
3rd deposit (start of year 3) earns interest for 1 year:	$3000(1.07)$

$A_3 = 3000(1.07) + 3000(1.07)^2 + 3000(1.07)^3 = 3000\left(1.07 + 1.07^2 + 1.07^3\right)$ as required.

(b) In the same way, $A_{25} = 3000\left(1.07 + 1.07^2 + \cdots + 1.07^{25}\right)$.

The bracket is a GP with $a = 1.07$, $r = 1.07$ and $n = 25$. Using $S_n = \dfrac{a\left(r^n - 1\right)}{r - 1}$ (Reference Sheet):

$$A_{25} = 3000 \times \frac{1.07\left(1.07^{25} - 1\right)}{1.07 - 1} = 203\,029.41\ldots$$

$\therefore$ The amount at the end of 25 years is \$203 029.41.

(c) Total deposited $= 25 \times 3000 = \$75\,000$. Interest $= 203\,029.41 - 75\,000 = \$128\,029.41$

Teaching notes

Write the deposits starting from the most recent one, so the powers increase: $1.07, 1.07^2, \ldots$. This makes the first term and ratio of the GP easy to read off.

Section D

Finance Applications of AP (Simple Interest and Constant Increases)

(i) Key words and vocabulary | (ii) 3 worked examples | 等差数列在金融中的应用（单利与固定增长）

D(i) Key words and vocabulary for AP finance 单利与固定增长

With simple interest the interest is calculated on the **original principal only**, so the **same dollar amount** is added every year: the yearly values form an AP. Any amount that "increases by \$100 per month" or "by \$2400 each year" is also an AP.

Key words and vocabulary	中文翻译 Chinese	What it tells you
simple interest	单利	Interest on the original principal only, the same every year: AP
earns 5% p.a. simple interest	按 5% 的年利率计算单利	Interest each year = 5% of the principal = d
flat-rate interest	固定利率（单利）	Another name for simple interest
the interest earned each year is the same	每年所得利息相同	Common difference d
increases by \$100 per month	每月增加 100 元	$d = 100$ (months)
salary increases by \$2400 each year; annual increment	工资每年增加 2400 元；年度加薪	$d = 2400$ (years)
starting salary	起薪	a, the first year's salary
saves \$50 more each week than the week before	每周比前一周多存 50 元	$d = 50$ (weeks)
repayments decrease by \$40 each month	每月还款减少 40 元	$d = -40$
total earnings; total savings	总收入；总储蓄	A sum, S_n
value at the end of year n	第 n 年年末的价值	One term, T_n
per annum (p.a.); per month; per week	每年；每月；每周	The time period that n counts
investment; deposit	投资；存款	The starting amount (principal)

Teaching notes

- The simple interest formula is not on the Advanced Reference Sheet. Find the interest for one year (for example, 5% of \$6000 = \$300). This is d, so the values form an AP and $T_n = a + (n-1)d$ (Reference Sheet) applies.
- Compare: simple interest *adds* the same amount each year (AP, linear growth). Compound interest *multiplies* by the same factor each year (GP, exponential growth).

D(ii) Worked examples: AP finance

Example D1 AP • finance Easy — Simple interest

Tom invests \$6000 at **5% p.a. simple interest**.

(a) Show that the values of the investment at the end of each year form an arithmetic sequence. (1 mark)

(b) Find the value of the investment at the end of 12 years. (1 mark)

(c) After how many years will the investment **first be worth at least** \$10 500? (2 marks)

Key words 关键词

5% p.a. simple interest （按 5% 的年利率计算单利）	⇒ the same interest every year: 5% of \$6000 = \$300 = d
first be worth at least （首次至少值）	⇒ solve; equality is allowed, so round up only if needed

NESA-style solution

(a) Simple interest is calculated on the original \$6000 only, so the interest each year is

$$5\% \times 6000 = 0.05 \times 6000 = \$300$$

The values at the end of each year are $6300, 6600, 6900, \ldots$. Each value is \$300 more than the previous one, so the values form an arithmetic sequence with $a = 6300$ and $d = 300$.

(b) Using $T_n = a + (n-1)d$ (Reference Sheet): $T_{12} = 6300 + 11 \times 300 = 9600$ $\therefore$ \$9600

(c) Solve $T_n = 10\,500$:

$$6300 + 300(n-1) = 10\,500 \implies 300(n-1) = 4200 \implies n - 1 = 14 \implies n = 15$$

At the end of 15 years the value is exactly \$10 500, which is "at least \$10 500".

$\therefore$ After 15 years.

Example D2 AP • finance Moderate — Salary with annual increments

Priya's **starting salary** is \$58 000. Her salary **increases by \$2400 each year**.

(a) Find her salary in the 10th year. (1 mark)

(b) Find her **total earnings** over the first 10 years. (1 mark)

(c) In which year will her salary **first exceed** \$90 000? (2 marks)

Key words 关键词

starting salary （起薪）	$\Rightarrow a = 58\,000$
increases by \$2400 each year （每年增加 2400 元）	$\Rightarrow$ AP with $d = 2400$
total earnings （总收入）	$\Rightarrow$ a sum, S_n
first exceed （首次超过）	$\Rightarrow$ solve, then round up

NESA-style solution

(a) The yearly salaries form an arithmetic sequence with $a = 58\,000$ and $d = 2400$.
Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{10} = 58\,000 + 9 \times 2400 = 79\,600 \qquad \therefore \$79\,600$$

(b) Using $S_n = \frac{n}{2}(a + l)$ (Reference Sheet) with $l = T_{10} = 79\,600$:

$$S_{10} = \frac{10}{2}(58\,000 + 79\,600) = 5 \times 137\,600 = 688\,000 \qquad \therefore \$688\,000$$

(c) Solve $T_n = 90\,000$:

$$58\,000 + 2400(n-1) = 90\,000 \quad \Longrightarrow \quad n - 1 = \frac{32\,000}{2400} = 13.33\ldots \quad \Longrightarrow \quad n = 14.33\ldots$$

Her salary increases every year, so it first exceeds \$90 000 in year 15.
Check: $T_{14} = 58\,000 + 13 \times 2400 = \$89\,200$; $T_{15} = 58\,000 + 14 \times 2400 = \$91\,600$.
$\therefore$ Year 15.

Example D3 AP • finance Challenging — Savings that increase each month

Leo saves \$200 in the first month. He then **increases the amount he saves by \$100 per month**.

(a) How much does he save in the 12th month? (1 mark)

(b) Show that his **total savings** after n months are $50n^2 + 150n$ dollars. (2 marks)

(c) After how many months will his total savings **first exceed** \$20 000? (2 marks)

Key words 关键词

increases … by \$100 per month （每月增加 100 元）	⇒ AP with $d = 100$
\$200 in the first month （第一个月存 200 元）	⇒ $a = 200$
total savings （总储蓄）	⇒ a sum, S_n
first exceed （首次超过）	⇒ solve a quadratic equation, then round up

NESA-style solution

(a) The monthly savings form an arithmetic sequence with $a = 200$ and $d = 100$.
Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{12} = 200 + 11 \times 100 = 1300 \qquad \therefore \$1300$$

(b) Using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_n = \frac{n}{2}[2(200) + (n-1)(100)] = \frac{n}{2}(100n + 300) = 50n^2 + 150n \quad \text{as required.}$$

(c) Solve $S_n = 20\,000$:

$$50n^2 + 150n = 20\,000$$
$$n^2 + 3n - 400 = 0$$
$$n = \frac{-3 \pm \sqrt{3^2 - 4(1)(-400)}}{2(1)} = \frac{-3 \pm \sqrt{1609}}{2} \qquad \text{(quadratic formula (Reference Sheet))}$$
$$n = 18.55\ldots \quad (n > 0)$$

His total savings increase every month, so they first exceed \$20 000 after 19 months.
Check: $S_{18} = 50(18)^2 + 150(18) = \$18\,900$; $S_{19} = 50(19)^2 + 150(19) = \$20\,900$.
$\therefore$ After 19 months.

Teaching notes

This is the AP version of a savings plan: the *deposit* grows by a fixed amount each month. Contrast it with Example C3, where a fixed deposit grows by compound interest (GP).

Section E

Practice Questions: Applications of AP and GP (Mixed)

10 HSC-style questions | Easy (1–3), Moderate (4–7), Challenging (8–10) | Total: 42 marks | 综合练习

These questions can be copied for students. All compound interest is compounded annually. Formulas must come from the NESA Reference Sheet. NESA-style solutions follow the questions.

Question 1 AP Easy — 3 marks

A factory produces 500 bicycles in its first week. Production increases by 25 bicycles each week.

(a) How many bicycles are produced in the 12th week? (1 mark)

(b) How many bicycles are produced altogether in the first 12 weeks? (2 marks)

Question 2 GP Easy — 3 marks

A video is shared 4 times in the first hour after it is posted. In each hour after that, the number of shares is 3 times the number in the previous hour.

(a) How many times is the video shared in the 6th hour? (1 mark)

(b) What is the total number of shares in the first 6 hours? (2 marks)

Question 3 GP Easy — 2 marks

A pattern of squares is drawn. The first square has a side length of 16 cm, and each square after that has a side length that is three-quarters of the side length of the previous square. The pattern continues indefinitely.

Find the total perimeter of all the squares. (2 marks)

Question 4 AP Moderate — 4 marks

Jordan does 20 push-ups on the first day of a fitness program and does 3 more push-ups each day than the day before.

(a) Show that the total number of push-ups Jordan does in the first n days is $\dfrac{n(3n+37)}{2}$. (2 marks)

(b) On which day will Jordan's total number of push-ups first exceed 1000? (2 marks)

Question 5 GP Moderate — 3 marks

The population of a country town is 12 000. Each year the population decreases by 3%.

(a) Find the population after 5 years, to the nearest whole number. (1 mark)

(b) After how many full years will the population first fall below 9000? (2 marks)

Question 6 AP • finance Moderate — 5 marks

Mai invests \$8000 at 4.5% p.a. simple interest.

(a) Show that the value of the investment at the end of n years is $(8000 + 360n)$ dollars. (2 marks)

(b) Find the value of the investment at the end of 7 years. (1 mark)

(c) At the end of which year will the investment first be worth more than \$12 000? (2 marks)

Question 7 AP & GP Moderate 7 marks

Two job offers each have a starting salary of \$60 000 per year.

Offer A: the salary increases by \$3000 each year.
Offer B: the salary increases by 4% each year.

(a) Find the salary in the 10th year for each offer. (2 marks)

(b) Find the total earnings over the first 10 years for each offer. (3 marks)

(c) Show that Offer B pays a higher salary than Offer A in the 13th year, but not in the 12th year. (2 marks)

Question 8 GP • finance Challenging 5 marks

At the start of each year, Sam invests \$4000 into an account that earns 5% p.a. interest, compounded annually.

(a) Show that the amount in the account at the end of n years is $A_n = 84\,000\,(1.05^n - 1)$. (2 marks)

(b) Find the amount in the account at the end of 20 years. (1 mark)

(c) At the end of which year will the amount in the account first exceed \$200 000? (2 marks)

Question 9 GP • finance Challenging 5 marks

Lena borrows \$50 000 at 6% p.a. interest, compounded annually. At the end of each year, interest is added to the amount owing and then Lena makes a repayment of $\$M$. Let A_n be the amount owing after the nth repayment.

(a) Show that $A_2 = 50\,000(1.06)^2 - M(1 + 1.06)$. (2 marks)

(b) Show that $A_n = 50\,000(1.06)^n - \dfrac{M\,(1.06^n - 1)}{0.06}$. (1 mark)

(c) Find the value of M, to the nearest cent, if the loan is repaid in exactly 8 years. (2 marks)

Question 10 GP Challenging 5 marks

A patient takes a 100 mg dose of a medicine every 24 hours. Just before each new dose, only 30% of the medicine that was in the body immediately after the previous dose remains.

(a) Show that the amount of medicine in the body immediately after the 3rd dose is 139 mg. (1 mark)

(b) Find the amount of medicine in the body immediately after the 10th dose, correct to 2 decimal places. (2 marks)

(c) Show that the amount of medicine in the body immediately after a dose will never exceed 143 mg. (2 marks)

Section E NESA-style solutions

Question 1 AP Easy — 3 marks

Key words: **increases by 25 bicycles each week** ⇒ AP with $a = 500$, $d = 25$; **altogether** ⇒ S_{12}.

(a) Using $T_n = a + (n-1)d$ (Reference Sheet): $T_{12} = 500 + 11 \times 25 = 775$ ∴ 775 bicycles.

(b) Using $S_n = \frac{n}{2}(a + l)$ (Reference Sheet): $S_{12} = \frac{12}{2}(500 + 775) = 6 \times 1275 = 7650$ ∴ 7650 bicycles.

Question 2 GP Easy — 3 marks

Key words: **3 times the number in the previous hour** ⇒ GP with $a = 4$, $r = 3$; **total** ⇒ S_6.

(a) Using $T_n = ar^{n-1}$ (Reference Sheet): $T_6 = 4 \times 3^5 = 972$ ∴ 972 shares.

(b) Using $S_n = \frac{a(r^n - 1)}{r - 1}$ (Reference Sheet): $S_6 = \frac{4(3^6 - 1)}{3 - 1} = \frac{4 \times 728}{2} = 1456$ ∴ 1456 shares.

Question 3 GP Easy — 2 marks

Key words: **three-quarters of the previous** ⇒ GP with $r = \frac{3}{4}$; **continues indefinitely** ⇒ limiting sum.

The perimeters are $4 \times 16 = 64$, $4 \times 12 = 48$, $4 \times 9 = 36$, …, a GP with $a = 64$ and $r = \frac{3}{4}$.

Since $|r| < 1$, using $S = \frac{a}{1 - r}$ (Reference Sheet): $S = \frac{64}{1 - \frac{3}{4}} = 256$ ∴ The total perimeter is 256 cm.

Question 4 AP Moderate — 4 marks

Key words: **3 more push-ups each day** ⇒ AP with $a = 20$, $d = 3$; **total … first exceed** ⇒ solve $S_n = 1000$, round up.

(a) Using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_n = \frac{n}{2}[2(20) + (n-1)(3)] = \frac{n}{2}(3n + 37) = \frac{n(3n + 37)}{2} \quad \text{as required.}$$

(b) Solve $\frac{n(3n + 37)}{2} = 1000$, that is, $3n^2 + 37n - 2000 = 0$. Using the quadratic formula (Reference Sheet):

$$n = \frac{-37 \pm \sqrt{37^2 - 4(3)(-2000)}}{2(3)} = \frac{-37 \pm \sqrt{25\,369}}{6} = 20.37\ldots \quad (n > 0)$$

Check: $S_{20} = \frac{20 \times 97}{2} = 970 \leq 1000$; $S_{21} = \frac{21 \times 100}{2} = 1050 > 1000$.

∴ Day 21.

Question 5 GP Moderate 3 marks

Key words: **decreases by 3%** ⇒ GP with $r = 0.97$; **first fall below** ⇒ logarithms, round up.

(a) The populations $12\,000,\ 12\,000 \times 0.97, \ldots$ form a GP with $a = 12\,000$ and $r = 0.97$. After n years is the $(n+1)$th term, $12\,000 \times 0.97^n$ (using $T_n = ar^{n-1}$ (Reference Sheet)).
After 5 years: $12\,000 \times 0.97^5 = 10\,304.8\ldots$ $\therefore$ 10 305 people.

(b) Solve $12\,000 \times 0.97^n = 9000$:

$$0.97^n = 0.75 \quad \Longrightarrow \quad n = \log_{0.97} 0.75 = \frac{\ln 0.75}{\ln 0.97} = 9.44\ldots \qquad \text{(change of base (Reference Sheet))}$$

The population decreases every year. Check: after 9 years ≈ 9123; after 10 years ≈ 8849.
$\therefore$ After 10 years.

Question 6 AP • finance Moderate 5 marks

Key words: **simple interest** ⇒ the same interest every year ⇒ AP.

(a) Interest each year $= 4.5\% \times 8000 = \$360$. The values at the end of each year are $8360, 8720, \ldots$, an AP with $a = 8360$ and $d = 360$.
Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_n = 8360 + 360(n-1) = 8000 + 360n \quad \text{as required.}$$

(b) $T_7 = 8000 + 360 \times 7 = 10\,520$ $\therefore$ \$10 520

(c) Solve $8000 + 360n = 12\,000$: $n = \dfrac{4000}{360} = 11.11\ldots$
Check: $T_{11} = \$11\,960$ (not more than \$12 000); $T_{12} = \$12\,320$.
$\therefore$ At the end of year 12.

Question 7 AP & GP Moderate 7 marks

Key words: **increases by \$3000 each year** ⇒ AP ($a = 60\,000$, $d = 3000$); **increases by 4% each year** ⇒ GP ($a = 60\,000$, $r = 1.04$).

(a) Offer A, using $T_n = a + (n-1)d$ (Reference Sheet): $T_{10} = 60\,000 + 9 \times 3000 = \$87\,000$
Offer B, using $T_n = ar^{n-1}$ (Reference Sheet): $T_{10} = 60\,000 \times 1.04^9 = \$85\,398.71$

(b) Offer A, using $S_n = \dfrac{n}{2}[2a + (n-1)d]$ (Reference Sheet): $S_{10} = \dfrac{10}{2}[2(60\,000) + 9(3000)] = 5 \times 147\,000 = \$735\,000$

Offer B, using $S_n = \dfrac{a(r^n - 1)}{r - 1}$ (Reference Sheet): $S_{10} = \dfrac{60\,000\left(1.04^{10} - 1\right)}{1.04 - 1} = \$720\,366.43$

(c) Year 12: A: $60\,000 + 11 \times 3000 = \$93\,000$; B: $60\,000 \times 1.04^{11} = \$92\,367.24$, which is less than A.
Year 13: A: $60\,000 + 12 \times 3000 = \$96\,000$; B: $60\,000 \times 1.04^{12} = \$96\,061.93$, which is more than A.
$\therefore$ Offer B pays more than Offer A in the 13th year, but not in the 12th year.

Question 8 GP • finance Challenging 5 marks

Key words: **at the start of each year** ⇒ each deposit earns interest for a different number of years; **5% p.a. compounded annually** ⇒ multiply by 1.05 each year.

(a) Using $A = P(1+r)^n$ (Reference Sheet) for each deposit: the first deposit earns interest for n years, the next for $n-1$ years, …, the last for 1 year.

$$A_n = 4000(1.05) + 4000(1.05)^2 + \cdots + 4000(1.05)^n = 4000\left(1.05 + 1.05^2 + \cdots + 1.05^n\right)$$

The bracket is a GP with $a = 1.05$, $r = 1.05$ and n terms. Using $S_n = \dfrac{a\left(r^n - 1\right)}{r - 1}$ (Reference Sheet):

$$A_n = 4000 \times \frac{1.05\left(1.05^n - 1\right)}{1.05 - 1} = 4000 \times 21\left(1.05^n - 1\right) = 84\,000\left(1.05^n - 1\right) \quad \text{as required.}$$

(b) $A_{20} = 84\,000\left(1.05^{20} - 1\right) = 138\,877.007\ldots \qquad \therefore\ \$138\,877.01$

(c) Solve $84\,000\left(1.05^n - 1\right) = 200\,000$:

$$1.05^n = 1 + \frac{200\,000}{84\,000} = 3.3809\ldots$$

$$n = \log_{1.05} 3.3809\ldots = \frac{\ln 3.3809\ldots}{\ln 1.05} = 24.96\ldots \qquad \text{(change of base (Reference Sheet))}$$

The amount increases each year. Check: $A_{24} \approx \$186\,908.40$; $A_{25} \approx \$200\,453.82$.
$\therefore$ At the end of year 25.

Question 9 GP • finance Challenging 5 marks

Key words: **6% p.a., compounded annually** ⇒ multiply by 1.06 each year; **repayment of $\$M$** ⇒ subtract M each year.

(a) After one year, interest is added and M is repaid: $A_1 = 50\,000(1.06) - M$

$$\begin{aligned} A_2 &= A_1(1.06) - M = [50\,000(1.06) - M](1.06) - M \\ &= 50\,000(1.06)^2 - 1.06M - M = 50\,000(1.06)^2 - M(1 + 1.06) \quad \text{as required.} \end{aligned}$$

(b) Continuing the pattern: $A_n = 50\,000(1.06)^n - M\left(1 + 1.06 + 1.06^2 + \cdots + 1.06^{n-1}\right)$

The bracket is a GP with $a = 1$, $r = 1.06$ and n terms. Using $S_n = \dfrac{a\left(r^n - 1\right)}{r - 1}$ (Reference Sheet):

$$A_n = 50\,000(1.06)^n - M \times \frac{1.06^n - 1}{1.06 - 1} = 50\,000(1.06)^n - \frac{M\left(1.06^n - 1\right)}{0.06} \quad \text{as required.}$$

(c) The loan is repaid when $A_8 = 0$:

$$50\,000(1.06)^8 = \frac{M\left(1.06^8 - 1\right)}{0.06} \quad \Longrightarrow \quad M = \frac{0.06 \times 50\,000(1.06)^8}{1.06^8 - 1} = 8051.797\ldots$$

$\therefore\ M = \$8051.80$

Question 10 GP Challenging 5 marks

Key words: **only 30% … remains** ⇒ multiply by 0.3; **never exceed** ⇒ limiting sum.

(a) After dose 1: 100 mg. Just before dose 2: $0.3 \times 100 = 30$, so after dose 2: $100 + 30 = 130$ mg.
Just before dose 3: $0.3 \times 130 = 39 = 30 + 9$, so after dose 3: $100 + 30 + 9 = 139$ mg as required.

(b) Immediately after the nth dose the amount is $100 + 100(0.3) + 100(0.3)^2 + \cdots + 100(0.3)^{n-1}$, a GP with $a = 100$ and $r = 0.3$. Using $S_n = \dfrac{a(1-r^n)}{1-r}$ (Reference Sheet):

$$S_{10} = \frac{100\left(1-0.3^{10}\right)}{1-0.3} = 142.856\ldots \qquad \therefore 142.86 \text{ mg}$$

(c) Since $|r| = 0.3 < 1$, the series has a limiting sum. Using $S = \dfrac{a}{1-r}$ (Reference Sheet):

$$S = \frac{100}{1-0.3} = 142.857\ldots \text{ mg}$$

The amount after each dose increases towards 142.857… mg but never reaches it. Since $142.857\ldots < 143$, the amount never exceeds 143 mg.

Answers at a glance

Q	Answers	Marks
1	(a) 775 bicycles (b) 7650 bicycles	3
2	(a) 972 shares (b) 1456 shares	3
3	256 cm	2
4	(a) shown (b) day 21	4
5	(a) 10 305 (b) after 10 years	3
6	(a) shown (b) \$10 520 (c) the end of year 12	5
7	(a) A: \$87 000; B: \$85 398.71 (b) A: \$735 000; B: \$720 366.43 (c) shown	7
8	(a) shown (b) \$138 877.01 (c) the end of year 25	5
9	(a) shown (b) shown (c) M = \$8051.80	5
10	(a) shown (b) 142.86 mg (c) shown (limiting sum = 142.857 … mg)	5
	Total	**42**

Vocabulary List: Applications of AP and GP

词汇表：等差数列与等比数列的应用 | English – 中文 – symbol

Sequences and series 数列与级数		
sequence	数列	
term	项	T_n
first term	首项	a
nth term; general term	第 n 项；通项	T_n
last term	末项	l
series	级数（各项之和）	
sum of the first n terms	前 n 项和	S_n
arithmetic sequence (AP)	等差数列	
common difference	公差	d
geometric sequence (GP)	等比数列	
common ratio	公比	r
limiting sum; sum to infinity	极限和；无穷项和	S
converge	收敛	$\|r\| < 1$
diverge	发散	
increase; decrease	增加；减少	
constant; fixed	常数；固定的	
successive; consecutive	相继的；连续的	
double; triple; halve	翻倍；变为三倍；减半	
percentage increase	百分比增长	
percentage decrease	百分比减少	
linear growth	线性增长	AP
exponential growth; decay	指数增长；指数衰减	GP
equation; inequality	方程；不等式	
logarithm	对数	log, ln
quadratic equation	二次方程	
exceed; at least; at most	超过；至少；至多	
round up; round down	向上取整；向下取整	
correct to 2 decimal places	精确到小数点后两位	

Finance 金融		
financial mathematics	金融数学	
principal	本金	P
interest	利息	
interest rate	利率	r
per annum (p.a.)	每年	
per month; per week	每月；每周	
simple interest	单利	AP
compound interest	复利	GP
compounded annually	按年复利	
amount; future value	本利和；终值	A
investment; invest	投资	
deposit; withdrawal	存款；取款	
balance	余额	
depreciation	折旧；贬值	
appreciation	增值	
salvage value; book value	残值；账面价值	
salary; wage	工资；薪水	
starting salary	起薪	a
annual increment (pay rise)	年度加薪	d
savings	储蓄	
annuity	年金	
superannuation	养老金（退休金）	
contribution	供款	
loan; borrow	贷款；借款	
repayment; instalment	还款；分期付款	M
reducing-balance loan	余额递减贷款	
amount owing	欠款金额	A_n
to the nearest cent; dollar	精确到分；精确到元	

Remember 记住： **add** the same amount each time (加上相同的数) ⇒ **AP** $T_n = a + (n-1)d$; **multiply** by the same number each time (乘以相同的数) ⇒ **GP** $T_n = ar^{n-1}$; compound interest (复利) ⇒ $A = P(1+r)^n$; "eventually" (最终) ⇒ $S = \dfrac{a}{1-r}$, $|r| < 1$.